

Cypress Test Framework: Multi- Environment & Suite-Based Execution

Narayanan K

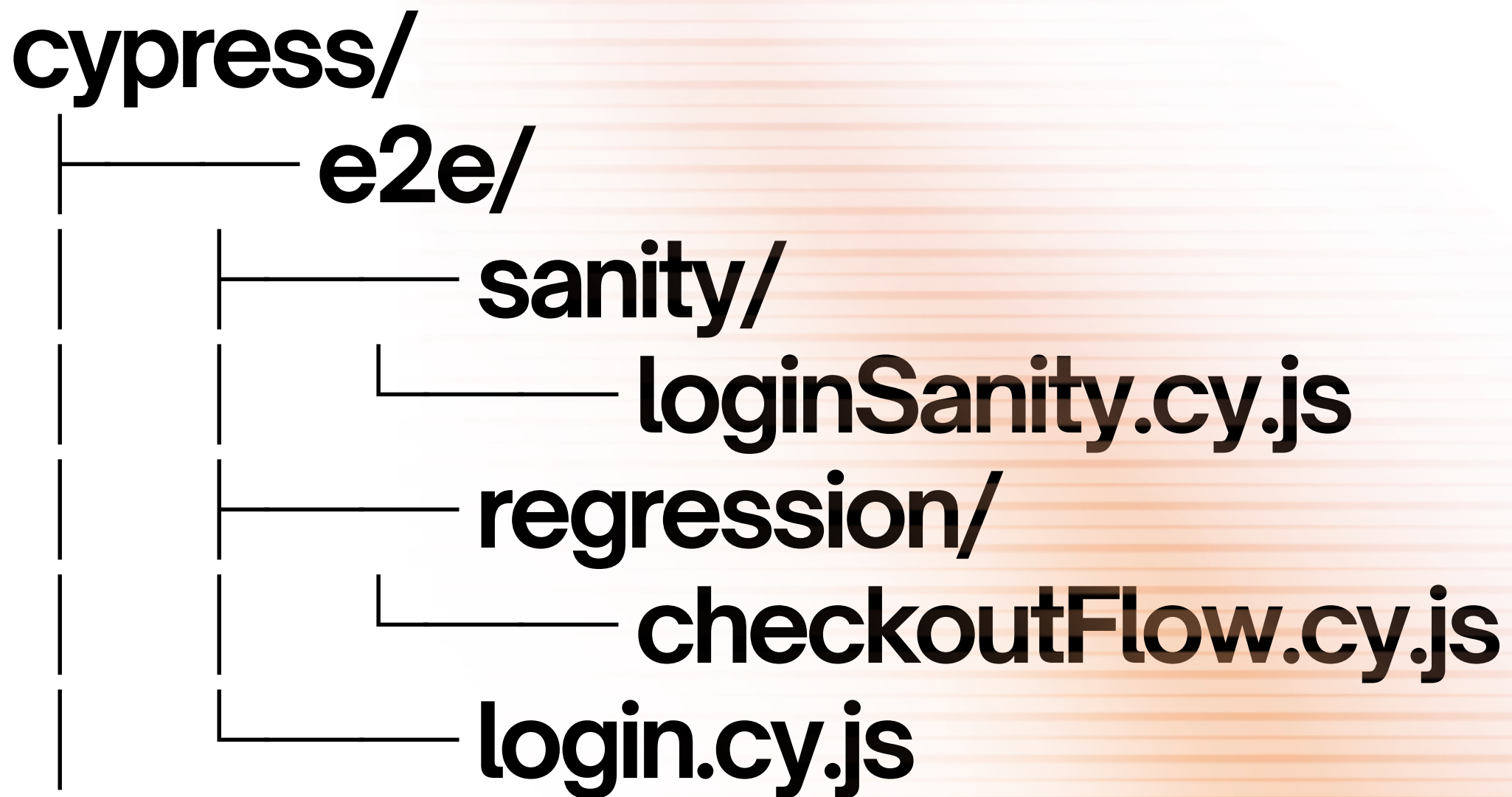


Goal of the Setup

To build a scalable Cypress automation framework that supports:

- Running tests in multiple environments (Dev, Staging, Prod)
- Creating and executing Sanity and Regression suites
- Organizing code for maintainability and flexibility

Step 1: Folder Structure



.env.dev

.env.staging

.env.prod

cypress.config.js

package.json

Step 2: Environment Setup with .env Files

.env.dev

```
CYPRESS_baseUrl=https://dev.yourapp.com  
CYPRESS_username=test_dev_user
```

.env.staging

```
CYPRESS_baseUrl=https://staging.yourapp.com  
CYPRESS_username=test_staging_user
```

.env.prod

```
CYPRESS_baseUrl=https://yourapp.com  
CYPRESS_username=real_user
```

Step 3: Configuring Cypress with Dotenv

```
const { defineConfig } = require('cypress');
const dotenv = require('dotenv');

module.exports = defineConfig({
  e2e: {
    baseUrl: process.env.CYPRESS_baseUrl,
    setupNodeEvents(on, config) {
      const envFile = `.${env}.${process.env.NODE_ENV || 'dev'}`;
      dotenv.config({ path: envFile });
      config.env.username = process.env.CYPRESS_username;
      return config;
    },
  },
});
```

Step 4: Writing Tests with Tags

Sanity Test: loginSanity.cy.js

```
describe('Sanity Suite – Login Test', { tags: ['sanity'] }, () => {  
  it('logs in with basic validation', () => {  
    cy.visit('/');  
    cy.get('#username').type(Cypress.env('username'));  
    cy.get('#password').type('test@123');  
    cy.get('#login').click();  
    cy.contains('Dashboard').should('exist');  
  });  
});
```

Regression Test: checkoutFlow.cy.js

```
describe('Regression Suite – Checkout Flow', { tags: ['regression'] }, () => {  
  it('completes checkout process', () => {  
    cy.visit('/products');  
    cy.get('.buy-now').first().click();  
    cy.get('#checkout').click();  
    cy.contains('Thank you for your purchase!').should('be.visible');  
  });  
});
```

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Step 5: Package Scripts for Easy Execution

```
"scripts": {  
  "test:dev": "NODE_ENV=dev cypress run",  
  "test:staging": "NODE_ENV=staging cypress run",  
  "test:prod": "NODE_ENV=prod cypress run",  
  
  "sanity": "NODE_ENV=dev cypress run --env grepTags=sanity",  
  "regression": "NODE_ENV=dev cypress run --env grepTags=regression",  
  
  "sanity:folder": "NODE_ENV=dev cypress run --spec 'cypress/e2e/sanity/*.cy.js'",  
  "regression:folder": "NODE_ENV=dev cypress run --spec 'cypress/e2e/regression/*.cy.js'",  
  
  "all": "NODE_ENV=dev cypress run --spec 'cypress/e2e/sanity/*.cy.js,cypress/e2e/regression/*.cy.js'",  
}
```

Step 6: How to Run

Command	Description
npm run test:dev	Run all tests in dev environment
npm run sanity	Run all sanity tests using tag
npm run regression	Run all regression tests using tag
npm run sanity:folder	Run all files under sanity/ folder
npm run all	Run both sanity and regression tests together